

Samarth Tiwari

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Education

Vellore Institute of Technology

B.Tech in Computer Science and Engineering (Cloud Computing and Automation)

Sep 2023 – May 2027

CGPA: 8.61/10

Technical Skills

Languages: Python, Java, JavaScript, SQL

Frontend: HTML5, CSS3

Systems & Backend: FastAPI, Microservices, WebSocket, Flask, REST APIs, Spring Boot

Cloud (AWS): Lambda, API Gateway, S3, EC2, CloudWatch, EventBridge, IAM, Bedrock, VPC

Databases: MongoDB, MySQL, DynamoDB

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Matplotlib

Tools & Practices: Docker, Kubernetes, Git, Postman, CI/CD, SDLC

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Serverless Computing

Experience

Python Developer – AI Intern

May 2025 – July 2025

ExaTip Technologies

Indore, MP

- Improved AI response reliability by analyzing user interactions, identifying retrieval gaps, and optimizing the knowledge base, resulting in more accurate responses.
- Built automated reporting workflows from database records, enabling faster access to business insights and operational metrics.
- Optimized AWS Lambda-based serverless workloads, improving application performance and resource utilization.
- Guided 4 interns on AWS cloud fundamentals and core services, supporting effective cloud adoption and technical upskilling.

Projects

Driver Drowsiness Detection - [GitHub](#)

Feb 2026 – Mar 2026

Real-Time Road-Safety AI System

Python, FastAPI, WebSocket, CNN, OpenCV, Docker

- Built a real-time CNN-based drowsiness detector (Conv2D 32→32→64 + Softmax) streaming webcam frames at 10 FPS over **WebSocket**, achieving sub-200 ms inference latency — cutting false-negative safety alerts by ~30% via confidence-threshold fallback logic.
- Containerized the full-stack app (**FastAPI** backend + Vercel frontend) with **Docker** and published to Docker Hub; reduced deployment setup from hours to under 2 minutes across any cloud or local environment.
- Boosted low-light detection accuracy by ~25% using **OpenCV Haar Cascade** ROI extraction with Histogram Equalization, ensuring reliable performance across real-world lighting conditions.

Packet Analyser - [GitHub](#)

Jan 2026 – Mar 2026

High-Performance Deep Packet Inspection (DPI) Engine

Python, Computer Networks

- Engineered a **multi-threaded DPI pipeline** with consistent hashing and lock-safe queues, enabling zero-race concurrent packet processing with accurate stateful flow tracking across high-volume traffic.
- Reduced traffic classification latency by ~40% via TLS SNI extraction and five-tuple flow modeling, enabling real-time application identification and automated policy enforcement at scale.

Sahayak AI - [GitHub](#)

Sept 2025 – Jan 2026

An AI-Powered Classroom Assistant for Indian Middle Schools

AWS Lambda, Bedrock, DynamoDB, S3, Python

- Automated end-to-end assignment evaluation pipeline, reducing manual grading effort by ~80% by processing Google Forms/Doc submissions via **Lambda** and **Bedrock**.
- Scaled serverless architecture handling structured/unstructured responses, enabling real-time grading and feedback delivery with **DynamoDB** storage and email notifications.
- Enabled data-driven decision making by aggregating **student performance data** and powering dashboard insights QuickSight for class-level and individual tracking.

Achievements & Certifications

- **AWS Certified Cloud Practitioner** - June 2026 (Score: 1000/1000)
- **IBM Professional Data Scientist**
- **Winner** – Nebula Nexus Hackathon 2025, MIT Jaipur